(Please write your Exam Roll No.) *Exam Roll No.....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: ES-102 ***Subject- Programming in 'C'***

Time: 3hours **Max. Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt **all** questions: **[5x4=20]**

a) Difference between while and do-while loop?
b) Define different types of sorting?
c) Define pre-processor and its usage in programming?
d) Explain the basic structure and compilation process of C program?
e) Difference between structure and union?

UNIT-I

Q2 a) What is a token? Explain the different types of tokens like keyword, identifiers, constants, strings with examples? **[5]**

b) Difference between flowchart and algorithm? Write an algorithm and flowchart to find the factorial of a number? **[5]**

Q3 a) Define Operator? Explain with example different types of operator arithmetic, logical, bitwise and ternary operator? **[6]**

b) Find the output of the following program: - **[4]**

```
Int main()
{Char srt1[20]= "Hello", Str2[20]= "World";
Printf("%s\n", strcpy(str2,strcat(str1,str2)));
Printf("s\n", strrev(str1));
Printf("s\n", strlen(str2));
return 0;}
```

UNIT-II

Q4 a) Define Recursion function? Write a C program to find the Fibonacci series using recursion? **[5]**

b) Define an array? Write the advantages of an array? Write a C program to transpose of amatrix? **[5]**

Q5 a) Explain call by value and call by reference with example? **[5]**

b) Define string? How it can be declared and initialized? Explain any five-string function with example? **[5]**

UNIT-III

Q6 a) Explain File Handling? What is the need for file handling? Discuss file access modes for text files: r, a, w, r+, a+, w+? **[5]**

b) What is the need of a structure? How will you access the members of a structure? Write a program to read and print the details of a person using Structure? **[5]**

Q7 a) Define Pointer with an example? How to access a variable through pointer? Write a C program to add two variables using Pointer? **[5]**

b) Define File Handling and explain its operations? **[5]**

P.T.O.